

Postdoc-Main Manuscript

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Abstract

Blah.

Background

- [1] Data harmonization entails coordinating a systematic effort to align the information contained in datasets collected by distributed teams of epidemiologists. These efforts to integrate the records collected during various discrete studies are motivated by a desire to establish larger integrated datasets bearing greater statistical power and that facilitate comparison across cohorts.
- [2] [Something about major firms and software stacks, the general growth of this domain. Make reference to Isabel Fortier’s work.]
- [3] However, epidemiologists must reckon with the diversity of minor variations in data collection procedures, as well as ethico-legal concerns relating to the sharing of individual health records pertaining to human research subjects across numerous institutional and regional jurisdictions. [Reference some of the work cited in my collaboration with Alex Bernier.]
- [4] Overall, harmonization requires significant coordination, planning, and governance work. It’s more than just a technical data alignment. [More detail on this here.]
- [5] These concerns were especially acute during the wake of the Covid-19 pandemic. [More detail on this here.]
- [6] [The main take-away from the background is that there was an initial severe under-estimation of the practical challenges involved in harmonization, especially in the realm of goal-setting and establishing common expectations; this study documents some of those challenges.]
- [7] This study investigates the social, technical, administrative, and epistemic factors that scaffold data-sharing initiatives in epidemiological research. It articulates the motivations for doing data harmonization, identifies how value is ascertained, and describes the strategies employed to achieve the desired goals — including perceived and actual challenges, setbacks, opportunities, realizations, and lessons learned.
- [8] It takes to heart the notions that data are media that facilitate communication across different research contexts, that data are created with specific intent, and that data are bounded by the social, practical, and material circumstances of their creation. In light of these facts, the study approaches data-sharing as a means of reconciling the varied circumstances of datasets’ creation — both among themselves, and in relation to contexts of reuse. It therefore frames data-sharing as efforts to foster a series of collaborative ties beyond a project’s original intended scope.
- [9] The project addresses the following research questions:
- What are the objectives of data-sharing initiatives, how were they established, and what progress has been made to achieve them?
 - What strategies do data-sharing initiatives employ to ensure they are able to meet their objectives, and how effective are they?

- What values underlie these strategies, and can they be linked with effective outcomes, such as the production of harmonized datasets and research deriving therefrom?

[10] The intent is to ascertain what actions specific strategies entail, the circumstances in which each is adopted, the value that they bring, and the trade-offs involved. In other words, the study articulates the collaborative experiences that underlie data-sharing activities as a series of converging situated perspectives.

Methods

Study Site

[11] The project examines the CITF Databank as an in-depth case study. The CITF was established in April 2020 to catalyze, support, fund and harmonize knowledge on SARS-CoV-2 immunity for federal, provincial, and territorial decision-makers. The Databank centralizes research data from CITF-funded studies, assembling an integrated dataset of over 165,000 individual participant health records from research studies performed across Canada, including a harmonized dataset for a subset of 40 studies. The Databank was led by a core team within the CITF Secretariat, supported by dedicated personnel within the Office of Sponsored Research at McGill University and experts in data governance and data harmonization. Its pan-Canadian scope, involving data sharing across multiple universities, research institutes, and governmental agencies in the context of a global public health emergency, makes it a uniquely rich setting for examining how large-scale health data sharing governance is coordinated in practice.

[12] The effort to coordinate this large-scale effort “on the fly” as the pandemic was surging and evolving is an interesting yet necessary case of contradiction against the wisdom that harmonization should be carefully planned and coordinated. As the paper demonstrates, planning did occur, but in a rather improvised manner that led to several unique challenges and opportunities.

Data

[13] The study draws from approximately ten semi-structured interviews with individuals who lead, support, or participate in the CITF Databank’s operations, including professional researchers, research trainees, and administrative and technical support staff. Interviews are oriented by the study’s goal to document processes of reconciling different stakeholders’ interests as they converge in the formation of a common data resource. Specifically, interviews focus on participants’ motives, the challenges they experience, how they envision success and failure, their perceptions of their own roles and the roles of other team members and stakeholders, the values that inform their decisions, how the technological apparatus they set up enables them to realize their goals and values, and ways in which they believe the work could be improved.

[14] Semi-structured interviews were held either in person or via video conference. They following a broad pattern of inquiry that traces (1) participants’ goals and perspectives; (2) their project’s missions, purposes, and motivations; and (3) the practices, procedures, and relationships that enable them to operationalize their personal and collective goals. Questions were tailored to each individual so that they could address responses in general terms and in relation to their own specific experiences.

[15] Transcripts were generated using a locally-hosted Whisper model with diarization capabilities using noScribe (Dröge 2026) and then manually edited to correct errors - primarily mis-spellings of proper names and acronyms, speaker label reassignments, and minor issues with accent interpretation. All data are collected and curated in full compliance with the McGill University’s research ethics board.

Methods

Results

2025-06-25-TMurphy.txt (4)

- [16] [71:78] Tanya: Okay. Um, so it's been a little bit nebulous. Yeah. Yeah. (.) Zack: Um, with reference to the different kinds of work you described, um, do yo Tanya: I'd say I like both the kind of the strategic and coordinating the work a Zack: Uh, why do you think that is? Tanya: It's a very different mindset. Zack: Yeah. Tanya: Uh, the data analysis or reviewing a large document is, is just, it's ver
- [17] [123:128] Tanya: Mm-hmm. Okay. (..) Zack: Um, do certain people or groups evoke a sense of antagonism or discomfort, Tanya: Um, the only thing has been some, some investigators who were really not, Zack: If you want to tell me after the name, after we stopped recording, you can Tanya: But, um, but that didn't happen much.
- [18] [141:146] Tanya: Um, for whatever reason, early on, I mean, fairly early on, I guess, in m Zack: Um, if you could run the CITF or CITF data bank differently, what would yo Tanya: I think it, I think the contracts officer and us, by extension, would hav Zack: I have a sense that David could smell it, you know, he sort of knows a lit Tanya: Yeah.

2025-07-02-DBuckeridge.txt (6)

- [19] [61:68] David: Well, I mean, the, so we had everybody from more basic scientists through Zack: For them or for others? David: Yeah. The negatives for them in the sense that if they shared their data, Zack: Was that the primary worry? That was one of the primary worries that was p David: Um, effort, I guess. I guess some people complained that, you know, this, Zack: So this is for clinicians and labs, right? Uh, the latter, I guess you men David: Yeah. They're not even labs necessarily. They're, they're separate entiti
- [20] [84:89] David: Yeah. Zack: But the risk of doing that... I would have thought that working with a net David: Yeah. I think the flip side of what I saw coming from them more was that Zack: Pooled model, what do you mean by that? David: When we said data sharing, we meant like, give us your data, we're puttin
- [21] [94:99] David: Okay. Zack: Um, or what this is some sort of framework, maybe a, a followup is do you David: Well, it was a mix and it wasn't, it was a lot of people doing it, right? Zack: Um, but, but majority, it was through funding. And then individual study o David: Yeah, they would have got funded and then, as a condition of funding, had
- [22] [110:115] David: Well, and I'm using open in a, you know, not purely technical sense, righ Zack: Right. It's still controlled. David: There's still, yeah, restricted access or controlled access. (...) But, Zack: Yeah. Yeah. Yeah. Okay. (..) With regards to sort of assembling a team of David: Well, the money, I guess the budget. (.....) Yeah. (.....) I don't
- [23] [166:171] David: I'd hope so. But I'm not sure. Like, my sense about the genesis of this w Zack: Yeah. David: And that was also, in my sense, a bit of the pushback from Quebec. Right. Zack: Right. Right. Yeah. Okay. That's a whole new wrinkle for you. I'm sorry. Y David: Oh, yeah. I mean, I think we had a great, like I said, great people. Ther

2025-07-11-TEvans.txt (6)

- [24] [79:86] Tim: You know, there is informed consent in every survey. Um, and, uh, I think w Zack: Um, when working with individual study leaders, so not necessarily CBS, or Tim: Um, no, uh, in the sense that we had very clear expectations on data sharin Zack: Um, with regards to risk aversion, can you maybe articulate what risks, wh Tim: I think some institutions said, uh, that they didn't trust data being analy Zack: Can

you clarify what kinds of institutions you're referring to? Is it univ Tim: Uh, HEMA-Québec was one where they were very reluctant to, uh, have their d

- [25] [93:100] Tim: And, uh, those agreements, I mean, there was a huge propensity to put it, t Zack: Um, would you, is there, um, is data sharing considered normal and ethical Tim: Actually not, I don't think. And I don't think it's, uh, part of the ethic Zack: Why do you think that is? Tim: I think it's because, uh, uh, epi has become, uh, uh, you know, an academic Zack: Mm-hmm. Tim: Um, which, uh, you know, they're in control of, but, uh, it's, uh, I think,
- [26] [104:109] Tim: Yeah, so, um, not really sure what you're asking. Zack: Sorry, can I rephrase that, then? Um, the motivations of individuals and o Tim: Yeah, no, I think, um, the academic culture is, is, um, very averse to data Zack: What makes the Canadian situation different? (..) Tim: Fourteen different jurisdictions, um, hyper-fragmented, um, uh, health syst

2025-10-15-OOxlade.txt (4)

- [27] [45:52] Olivia: Research assistants. Or research associates, actually, there was both. Zack: Right. Olivia: Um, so he ran those. So that wasn't part of my, um, responsibility. Um, Zack: Haha! You did, or you didn't? Olivia: You didn't. But you know, some of the communications ones, you're like, Zack: Yeah. Olivia: Yeah. I get this from talking about other people. I get the sense it was
- [28] [88:93] Olivia: Well, I think it was just a different, like same issue, but different at Zack: I mean, that stress alone is enough to inhibit someone from actually wanti Olivia: Right, right. Exactly. I mean, I think like people who were having those Zack: So you don't have to name names or anything, but I'm curious about two thi Olivia: Oh, very clear.
- [29] [128:133] Olivia: The grind, yeah. Zack: Um, were there any like a really, I mean, sometimes you really remember li Olivia: Yeah. Not that I wasn't really satisfied, but it wasn't like, you know, Zack: Or was it like an output that really was meaningful that really changed th Olivia: Well, probably once we knew, um, that the CITF was closing and the CITF

2026-01-14-SKhezri.txt (1)

- [30] [181:186] Sadun: Yeah. Zack: Uh, is that a factor maybe as well? Sadun: Uh, principal normally, uh, they try to, uh, to find and connect our, our Zack: Yeah. Sadun: Yeah. It's, it's, I think is my, is my mind. It's dependent the benefit o

2026-01-29-BMazer.txt (1)

- [31] [234:239] Bruce: So you have to have people who are willing. Zack: Yeah. Bruce: And when you then get lawyers and politicians involved, there's a lot of Zack: Yeah. Yeah, yeah, yeah. Yeah. Phenomenal opportunity. I just want to take Bruce: Sure. I hope this is useful.

Discussion

Conclusions

References

Dröge, Kai. 2026. "noScribe." <https://github.com/kaixxx/noScribe>.

List of Abbreviations

CIHR Canadian Institutes of Health Research;

CITF COVID-19 Immunity Task Force

DSA Data Sharing Agreement

FAIR Findable, Accessible, Interoperable, Reusable

LLM Large Language Model

MCHI McGill Clinical and Health Informatics

Ethics Approval and Consent to Participate

Consent for Publication

Availability of Data and Materials

Competing Interests

[32] The authors declare no competing interests.

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Author Contributions

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Authors' Bios